Construction Permit Timelines in San Francisco

Advanced Data Science end term project pre submission summary

1 Project and group Group no. _____ Members and student IDs _____
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2 Stakeholder and problem The San Francisco permitting operations team and building applicants need realistic timeline expectations and evidence for workload planning. We will identify which permit categories and property characteristics are associated with longer filing-to-issuance intervals, and where targeted process investigation is warranted. These intervals include both agency and applicant time.

3 Research questions and hypotheses H1: Issuance timelines differ by permit type. H2: Older buildings have longer timelines within comparable permit types and filing years. H3: Higher estimated-cost quartiles have longer timelines within those same groups. Each null hypothesis is no association or group difference. These are exploratory hypotheses, not causal claims.

4 Data sources and verified scope Authentic government records downloaded on 23 October 2026 IST. No Kaggle or synthetic observations. The prepared SSOT contains 154,689 unique permits and 95 columns: 55 permit, 18 assessor, 2 BLS and 20 derived or quality-control fields.

Official source and link	Coverage and rows	Key variables
SF Building Inspection data.sf.gov/d/i98e-djp9	Filed 2019-2024 154,689 permits	Permit type, filing/approval/issue dates, status, estimated and revised cost, units, stories, parcel
SF Assessor-Recorder data.sf.gov/d/wv5m-vpq2	Rolls 2017-2022 92,364 matched parcel-years	Built year, property area, property class, stories, units and zoning; extracted from 1,266,295 roll rows
US Bureau of Labor Statistics bls.gov/cpi/data.htm	2019-2024 72 monthly observations	CPI-U series CUUR0000SA0; broad purchasing-power normalization of estimated USD amounts

5 Integration plan and quality Keep primary_address_flag = Y and one row per permit number. Left-join on text block + lot + assessor roll year = filing year minus 2, avoiding later fiscal periods; match rate is 97.86%. Join CPI by filing month, with 100% coverage. Validate many-to-one joins and unchanged row counts. Preserve missing data and valid extremes; flag 18 negative issuance intervals. Record URLs, timestamps, hashes and field definitions. Units: calendar days, USD and square feet.

6 Methods and BADIR Business: define timeline and workload KPIs. Analysis plan: specify the three hypotheses and comparisons. Data: reproducible ETL, SQL aggregation and feature engineering. Insights: counts, medians, percentiles, line/bar/box plots, effect sizes and hypothesis tests. Recommendations: prioritise process reviews and applicant guidance. Apply relevant handout methods; add a small gradient-boosting benchmark only if prediction improves the decision, with chronological validation.

7 Validation and decision value Compare within permit type and year. Report group sizes, 95% confidence intervals and practical differences; use parcel-clustered resampling and Holm-adjusted tests at 5%. Check a common 365-day issuance window, annual stability and unmatched-property sensitivity. Do not interpret unissued/cancelled records as pending backlog. Use findings to select categories for investigation; no staffing savings are claimed without review-time data.

8 Limitations and next steps Snapshot outcomes and lagged property attributes may be incomplete or revised. Estimated costs are not actual spending; CPI is not a construction-cost index. Observational patterns do not establish causes. Next: verify records and missingness; produce stratified graphs/tests; assess hypotheses as supported, unsupported or inconclusive; prepare recommendations and the required public GitHub repository. Confirm final method scope and any deviation with the instructor.